Emily E. Costello

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EDUCATION & EMPLOYMENT

2023 – Present **PhD Candidate in Astrophysics** | Liverpool John Moores University & Liv.INNO Centre for Doctoral Training
Thesis Title: *How do Supermassive Black Holes affect their Host Haloes?*
Thesis Supervisors: Prof Ian G. McCarthy, Dr Jaime Salcido, Prof Robert A. Crain

Jul. 2026 – Jan. 2027 **Data Scientist** | Guy's and St Thomas' NHS Foundation Trust, Evelina Children's Hospital, KCL Department of Women and Young People's Health
Project Title: *Estimating the Unmet Need and Modelling the Impact of Scaled-Up Proactive Asthma Care in Children*
Project Supervisor: Prof Ingrid Wolfe

Oct. 2019 – Jul. 2023 **Astrophysics MPhys** | Liverpool John Moores University & University of Liverpool
Thesis Title: *Using Cosmological Simulations to Investigate the Impact of Black Hole Populations on Galaxy Groups and Clusters*
First Class Honours in Astrophysics. Thesis Supervisors: Prof Ian G. McCarthy, Dr Jaime Salcido

Jul. 2022 – Sep. 2022 **Nuclear Physics Summer Internship** | NuSec & University of Liverpool
Project Title: *Characterisation of Cadmium Zinc Telluride Crystals for Improved Position Resolution and Defect Correction in Nuclear Imaging and Security*
Project Supervisor: Dr Ellis J. Rintoul

RESEARCH

Publications

2026 **Measuring the tSZ Effect in FLAMINGO** First author
Emily E. Costello, Ian G. McCarthy, et al.
In prep

2026 **The Flamboyance project: an exploration of how baryonic feedback and cosmology conspire to shape large-scale structure**
Ian G. McCarthy, . . . , **Emily E. Costello**, et al.
In prep

2026 **Different halo gas reservoirs, similar galaxy populations: testing feedback with the FLAMINGO simulations**
Daudi T. Mazengo, Paola Popesso, Ilaria Marini, Joop Schaye, . . . , **Emily E. Costello**, et al.
In prep

2026 **Bird-brained galaxies: Synthetic unWISE galaxy catalogues from the FLAMINGO simulations**
Jonah T. Conley, Ian G. McCarthy, Tianyi Yang, . . . , **Emily E. Costello**, et al.
Submitted to MNRAS

2026 **FLAMINGO: The thermal history of the Universe as a probe for cosmology and galaxy formation**
Jaime Salcido, Tianyi Yang, Ian G. McCarthy, **Emily E. Costello**, et al.
arXiv:2605.11083

2026 **The FLAMINGO Simulations Data Release**
John C. Helly, Robert J. McGibbon, . . . , **Emily E. Costello**, et al.
doi:10.1016/j.ascom.2026.101159 | arXiv:2604.24324

2025 **FLAMINGO: Tracing the co-evolution of hot gas and black holes in galaxy groups and clusters** First author
Emily E. Costello, Ian G. McCarthy, Jaime Salcido, John C. Helly, Robert J. McGibbon, Matthieu Schaller, Joop Schaye
doi:10.1093/mnras/stag945 | arXiv:2510.17980

Seminars & Conferences

Jul. 2026 **International HPC Summer School** | Perth, Australia *Poster Presentation*

Jun. 2026 **UoM-LJMU-LU Extragalactic Meeting** | University of Manchester *Contributed Talk*

May 2026 **Baryons @ Princeton Workshop** | Princeton University, US *Invited Talk*

Mar. 2026 **Friday Lunchtime Astronomy Talk** | Durham University, UK *Invited Talk*

Jan. 2026 **Galaxies & Cosmology Theory Group** | Max Planck Institute for Astronomy (online) *Invited Talk*

Oct. 2025 **Multi-Phase, Multi-Temperature and Complex** | Sardinia, Italy *Contributed Talk*

Sep. 2025 **Prospero@5 Symposium** | Liverpool John Moores University, UK *Invited Talk*

Jul. 2025 **Cosmic Ecosystems** | Perimeter Institute for Theoretical Physics, Canada *Contributed Flash Talk*

Jan. 2025 **Virgo Consortium Collaboration Meeting** | University of Sussex, UK *Contributed Talk*

Sep. 2024 **Galaxy Clusters Journal Club** | University of Michigan, USA (online) *Invited Seminar*

Sep. 2024 **SWIFTcon Collaboration Meeting** | University of Leiden, The Netherlands *Contributed Talk*

Jul. 2024 **Liv.INNO Showcase Event** | University of Liverpool, UK *Poster Presentation*

Jul. 2024 **National Astronomy Meeting** | University of Hull, UK *Contributed Talk*

Jun. 2024 **Galaxy Groups in the Era of eROSITA & Euclid** | Centre for Astro., Sesto, Italy *Contributed Talk*

May 2024 **Alan Turing PhD Networking Day** | University of Liverpool, UK *Contributed Talk*

Sep. 2022 **Nuclear Security Science Network Conference** | Institute of Physics, UK *Poster Presentation*

Workshops, Summer Schools & Training

Jul. 2026 **International HPC Summer School** | Perth, Australia

Sep. 2025 **Fundamentals of GPU Accelerated Computing with Modern CUDA C++** | Nvidia (online)

Sep. 2024 **STFC Advanced Cosmology & Extragalactic School** | Sussex, UK

Oct. 2023 **DiRAC Foundations of HPC Skills Course** | (online)

Oct. 2023 **Liv.INNO Data Science Skills School** | Liverpool, UK

Sep. 2023 **Introductory Astronomy Summer School** | Hull, UK

AWARDS

2023 **Isaac Roberts Prize for Best Astrophysics Student** | University of Liverpool

2019 **Dr Joanne Hamilton Prize for Outstanding Contributions to Science** | Archbishop Blanch Sixth Form

TEACHING

University of Liverpool & Liverpool John Moores University: Graduate Teaching Assistant

2024, 2025, 2026 **PHYS374: Relativity and Cosmology** *Problems Classes*

2024 **PHYS216: Practical Astrophysics I** *Problems Classes*

2024 & 2025 **PHYS400: Practical Astrophysics II** *Field Trip Supervision*

University of Liverpool: Work Experience Week Project Supervisor

2025 & 2026 **Work Experience Week** *Project Leading & Supervision*

Liverpool John Moores University: The Schools' Observatory Outreach Assistant

2021 & 2023 **Work Experience Week** *Project Supervision*

ACTIVITIES

2023 – Present **Astrophysics Book Club Lead Organiser**
An inclusive, graduate-led club open to staff and students at every career stage across the Astrophysics Research Institute. The choice of book rotates between members, so people of different backgrounds and seniority can shape the reading list, which ensures for a diverse and inclusive selection of literature.

2024 – 2026 **Extragalactic & HPC Journal Club Lead Organiser**
The graduate-led journal club, designed to foster discussions among staff, students, and visitors, focuses on cutting-edge topics in extragalactic astrophysics and the latest advancements in high-performance computing relevant to the field.

2021 – 2023 **President / Treasurer of University of Liverpool Book Society**
Grew the society from around 100 to over 500 members, making it one of the largest in the University of Liverpool Guild of Students, through monthly events and discussions. As Treasurer, built links with local bookshops and secured member discounts. Established a Diversity Secretary role on the 8-person committee to ensure diverse reading lists, and ran free, accessible events to widen participation.

2021 – 2023 **Peer-Assisted Learning Lead Organiser**
Led a department-wide scheme with the aim of supporting struggling and/or disadvantaged first-year students with their learning and transition to university. Recruited and trained student mentors, and led sessions offering advice on university life and help with physics problems.

2020 – 2023 **Course Representative and Co-Chair of SSLC**
Represented students on the Student–Staff Liaison Committee, turning student feedback into constructive proposals for staff that led to many smaller changes across the department. Led an enhancement project that standardised formulae sheets across physics exams and increased the credit weighting of BSc and MPhys research projects; both changes remain in place since my graduation.